ASSIGNMENT NO: 11

Title: Write a C++ program to implement class and function templates.

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Class:SY1 PRN No:B25ET1013 Batch:A

```
#include <iostream>
using namespace std;
template <class T>
class Calculator
{
private:
T num1, num2;
public:
Calculator(T n1, T n2)
{
num1 = n1; num2 = n2;
}
void displayResult()
{
cout << "Numbers are: " << num1 << " and " << num2 << "." << endl;
cout << "Addition is: " << add() << endl;
cout << "Subtraction is: " << subtract() << endl;
cout << "Product is: " << multiply() << endl;
cout << "Division is: " << divide() << endl;
}
```

```
T add()

{
return num1 + num2;
}

T subtract()
{
return num1 - num2;
}

T multiply()
{
return num1 * num2;
}

T divide()
{
return num1 / num2;
}
};

int main()
{
Calculator<int> intCalc(2, 1);// Object with parameterized constructors
Calculator<float> floatCalc(2.4, 1.2);
cout << "Int results:" << endl;
```

```
intCalc.displayResult();

cout << endl << "Float results:" << endl;

floatCalc.displayResult();


return 0;
}
```

Output:

Output

```
Int results:
Numbers are: 2 and 1.
Addition is: 3
Subtraction is: 1
Product is: 2
Division is: 2

Float results:
Numbers are: 2.4 and 1.2.
Addition is: 3.6
Subtraction is: 1.2
Product is: 2.88
Division is: 2
```